Paste Key:

Σ, ä, ƀ , ċ , ∈ , ≡ , ⨁, α, ϕ,ℝ,

HW: DH and DLP

CYB-402

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1.

1.1 Z^*_5

All integers from 1 to 4 that are coprime with 5, so elements are {1,2,3,4}

$a^k \equiv 1 \pmod 5$ = order with smallest positive integer k

a	ord(a)
1	1
2	4
3	4
4	2

a = 1

$1^1 = 1 \rightarrow 1^1 \equiv 1 \pmod 5$

ord(1) = 1

a = 2

$2^1 = 2 \rightarrow 2 \pmod 5 = 2 \neq 1$

$2^2 = 4 \rightarrow 4 \pmod 5 = 4 \neq 1$

$2^3 = 8 \rightarrow 3 \pmod 5 = 3 \neq 1$

$2^4 = 2 * 2^3 = 2 * 8 = 16 \pmod 5 \equiv 1 \pmod 5$

$2^4 \equiv 1 \pmod 5$, ord(2) = 4

a = 3

$3^1 = 3 \rightarrow 3 \pmod 5 = 3 \neq 1$

$3^2 = 9 \rightarrow 4 \pmod 5 = 4 \neq 1$

$3^3 = 3 * 3^2 = 3 * 9 = 27 \rightarrow 2 \pmod 5 = 2 \neq 1$

$3^4 = 3 * 3^3 = 3 * 27 = 81 \pmod 5) \equiv 1 \pmod 5$

$3^4 \equiv 1 \pmod 5$, ord(3) = 4

a = 4

$4^1 = 4$ -> 4 (mod 5) = 4 != 1

$4^2 = 16$ -> 1 (mod 5)

$4^2 \equiv 1 \pmod 5$, ord(4) = 2

1.2 Z^*_{11}

All integers from 1 to 10 that are coprime to 11, so elements are {1,2,3,4,5,6,7,8,9,10}

$a^k \equiv 1 \pmod{11}$ = order with smallest positive integer k

Since 11 is prime, $|Z^*_{11}| = 100$, the order of any element must divide 10. Orders are 1,2,5,10.

a	ord(a)
1	1
2	10
3	5
4	5
5	5
6	10
7	10
8	10
9	5
10	2

a = 1

$1^1 = 1$ -> $1^1 \equiv 1 \pmod{11}$

ord(1) = 1

a = 2

$2^1 = 2$ -> 2 (mod 11) = 2 != 1

$2^2 = 4$ -> 4 (mod 11) = 4 != 1

$2^3 = 8$ -> 8 (mod 11) = 8 != 1

$2^4 = 2 * 2^3 = 2 * 8 = 16 \pmod{11} \equiv 5 \pmod 5$

$2^5 = 2 * 2^4 = 2 * 16 = 32 \pmod{11} \equiv 10 \pmod 5$

$2^6 = 2 * 2^5 = 2 * 32 = 64 \pmod{11} \equiv 9 \pmod 5$

$2^7 = 2 * 2^6 = 2 * 64 = 128 \pmod{11} \equiv 7 \pmod 5$

$2^8 = 2 * 2^7 = 2 * 128 = 256 \pmod{11} \equiv 3 \pmod 5$

$2^9 = 2 * 2^8 = 2 * 256 = 512 \pmod{11} \equiv 6 \pmod 5$

$2^{10} = 2 * 2^9 = 2 * 512 = 1024 \pmod{11} \equiv 1 \pmod 5$

$2^{10} \equiv 1 \pmod{11}$, ord(2) = 10

a = 3

$3^1 = 3$ -> 3 (mod 11) = 3 != 1

$3^2 = 9$ -> 4 (mod 11) = 9 != 1

$3^3 = 3 * 3^2 = 3 * 9 = 27$ -> 5 (mod 11) = 5 != 1

$3^4 = 3 * 3^3 = 3 * 27 = 81 \pmod{11} \equiv 4 \pmod{11}$

$3^5 = 3 * 3^4 = 3 * 81 = 243 \pmod{11} \equiv 1 \pmod{11}$

$3^5 \equiv 1 \pmod{11}$, ord(3) = 5

a = 4

$4^1 = 4$ -> 4 (mod 5) = 4 != 1

$4^2 = 16$ -> 5 (mod 5) != 1

$4^3 = 4 * 4^2 = 4 * 16 = 64$ -> 9 (mod 5)

$4^4 = 4 * 4^3 = 4 * 64 = 256$ -> 3 (mod 5)

$4^5 = 4 * 4^4 = 4 * 256 = 1024$ -> 1 (mod 5)

$4^5 \equiv 1 \pmod{11}$, ord(4) = 5

a = 5

$5^1 = 5$

$5^2 = 25$ -> 3 (mod 5) != 1

$5^3 = 5 * 5^2 = 5 * 25 = 125$ -> 4 (mod 5)

$5^4 = 5 * 5^3 = 5 * 125 = 625$ -> 9 (mod 5)

$5^5 = 5 * 5^4 = 5 * 625 = 3125$ -> 1 (mod 5)

$5^5 \equiv 1 \pmod{11}$, ord(5) = 5

a = 6

$6^1 = 6$

$6^2 = 36$ -> 3 (mod 11) = 3 != 1

$6^3 = 6 * 6^2 = 6 * 36 = 216$ -> 7 (mod 11)

$6^4 = 6 * 6^3 = 6 * 216 = 1296 \pmod{11} \equiv 9 \pmod{5}$

$6^5 = 6 * 6^4 = 2 * 1296 = 7776 \pmod{11} \equiv 9 \pmod{5}$

$6^6 = 6^4 * 6^2 = 9 * 3 = 27 \pmod{11} \equiv 5 \pmod{5}$

$6^7 = 6 * 6^6 = 6 * 5 = 30 \pmod{11} \equiv 8 \pmod{5}$

$6^8 = 6 * 6^7 = 6 * 8 = 48 \pmod{11} \equiv 4 \pmod{5}$

$6^9 = 6 * 6^8 = 6 * 4 = 24 \pmod{11} \equiv 2 \pmod{5}$

$6^{10} = 6 * 6^9 = 6 * 2 = 12 \pmod{11} \equiv 1 \pmod{5}$

$6^{10} \equiv 1 \pmod{11}$, ord(6) = 10

a = 7

$7^1 = 7$

$7^2 = 49$ -> 5 (mod 11) = 5 != 1

$7^3 = 7 * 7^2 = 7 * 49 = 343$ -> 2 (mod 11)

$7^4 = 7 * 7^3 = 7 * 2 = 14 \pmod{11} \equiv 3 \pmod{5}$

$7^5 = 7 * 7^4 = 7 * 3 = 21 \pmod{11} \equiv 10 \pmod{5}$

$7^6 = 7 * 7^5 = 7 * 10 = 70 \pmod{11} \equiv 4 \pmod{5}$

$7^7 = 7 * 7^6 = 7 * 4 = 28 \pmod{11} \equiv 6 \pmod{5}$

$7^8 = 7 * 7^7 = 7 * 6 = 42 \pmod{11} \equiv 9 \pmod{5}$

$7^9 = 7 * 7^8 = 7 * 9 = 63 \pmod{11} \equiv 8 \pmod{5}$

$7^{10} = 7 * 7^9 = 7 * 8 = 56 \pmod{11} \equiv 1 \pmod{5}$

$7^{10} \equiv 1 \pmod{11}$, ord(7) = 10

a = 8

$8^1 = 8$

$8^2 = 64$ -> 9 (mod 11) = 9 != 1

$8^3 = 8 * 8^2 = 8 * 9 = 72$ -> 6 (mod 11)

$8^4 = 8 * 8^3 = 8 * 6 = 48 \pmod{11} \equiv 4 \pmod 5$

$8^5 = 8 * 8^4 = 8 * 4 = 32 \pmod{11} \equiv 10 \pmod 5$

$8^6 = 8 * 8^5 = 8 * 10 = 80 \pmod{11} \equiv 3 \pmod 5$

$8^7 = 8 * 8^6 = 8 * 4 = 24 \pmod{11} \equiv 2 \pmod 5$

$8^8 = 8 * 8^7 = 8 * 6 = 16 \pmod{11} \equiv 5 \pmod 5$

$8^9 = 8 * 8^8 = 8 * 9 = 40 \pmod{11} \equiv 7 \pmod 5$

$8^{10} = 8 * 8^9 = 8 * 8 = 56 \pmod{11} \equiv 1 \pmod 5$

$8^{10} \equiv 1 \pmod{11}$, ord(8) = 10

a = 9

$9^1 = 5$

$9^2 = 81$ -> 4 (mod 5) != 1

$9^3 = 9 * 9^2 = 9 * 4 = 36$ -> 3 (mod 5)

$9^4 = 9 * 9^3 = 9 * 3 = 27$ -> 5 (mod 5)

$9^5 = 9 * 9^4 = 9 * 5 = 45$ -> 1 (mod 5)

$9^5 \equiv 1 \pmod{11}$, ord(9) = 5

a = 10

$10^1 = 10$

$10^2 = 100$ -> 4 (mod 5) != 1

$10^2 \equiv (-1)^2 = 1 \pmod{11}$

ord(10) = 2

2.

2.1 For $Z^*_{5,}$ p = 5 and listed early $Z^*_5 = \{1,2,3,4\}$, so it has 4 elements. For $Z^*_{11,}$ p = 11 and listed early $Z^*_{11} = \{1,2,3,4,5,6,7,8,9,10\}$ so it has 10 elements.

2.2

For $|Z^*_5| = 4$, orders found are ord(1) = 1, ord(2) = 4, ord(3) = 4, ord(4) = 2

Check if each order divides 4:

1	4	yes
4	4	yes
4	4	yes
2	4	yes

So yes, every element order in Z^*_5 divides 4.

For $|Z^*_{11}| = 10$, orders found are ord(1) = 1, ord(2) = 10, ord(3) = 5, ord(4) = 5, ord(5) = 5, ord(6) = 10, ord(7) = 10, ord(8) = 10, ord(9) = 5, ord(10) = 2

Check if each order divides 10:

1	10	yes
10	10	yes
5	10	yes
5	10	yes
5	10	yes
10	10	yes
10	10	yes
10	10	yes
5	10	yes
2	10	yes

So yes, every element order in Z^*_{11} divides 10.

2.3 For Z^*_5, primitive elements are with order 4, so {2,3}

For Z^*_{11}, primitive elements are with order 10, so {2,6,7,8}

2.4 For Z^*_5, $\varphi(|Z^*_5|) = \varphi(4)$

$4 = 2^2$, $\varphi(4) = 2^2 - 2^0 = 4 - 2 = 2$

For Z^*_{11}, $\varphi(|Z^*_{11}|) = \varphi(10)$

$= (2^1 - 2^0)(5^1 - 5^0) = 4*1 = 4$

3.1 Since 47 is prime, $(|Z^*_{47}|) = 47 - 1 = 46$. The order of any element must divide the order of the group, so any element order must be a divisor of 46.

$46 = 2 * 23$

Positive divisors of 46: 1,2,23,46.

So, the possible element order in Z^*_{47} are 1,2,23,46

3.2 Primitive element = element of max order. Order equal to $|Z^*_{47}|$

$\Phi(46)$

$46 = 2 * 23$

$\Phi(46) = 46(1 - ½)(1 - 1/23) = 46 * (1/2) * (22/23) = 23 * (22/23) = 22$, so the number of primitive elements in Z^*_{47} is 22.

4.

4.1 $p = 4967$, Group Z^*_{4967}.

$| Z^*_{4967} | = 4967 - 1 = 4966$

$4966 / 2 = 2483$ so, $4966 = 2 * 2483$

Check

$7 * 355 = 2485$,

$11 * 225 = 2475$,

$11 * 8 = 2486$,

$13 * 190 = 2470$,

$13 * 191 = 2483$.

$4966 = 2 * 13 * 191$

So, $\Phi(4966) = (2^1 - 2^0)(13^1 - 13^0)(191^1 - 191^0)$

$= 1 * 12 * 190 = 12 * 190 = 2280$

There are 2280 generators in Z^*_{4967}

4.2 Total elements is 4966, generators is 2280

So $2280/4966 = .4591 = 45.9\%$ chance of a randomly chosen element being a generator.

4.3 Because in actual practice p is extremely large so $Z^*_p = p - 1$ is enormous. In order to verify that the α generates the whole group by brute force it would need up to $p - 1$ exponentiation. Which is computationally just not practical and infeasible. So, when we're dealing with these small p it makes this random search actually feasible.

5. $p = 467$, $\alpha = 2$

Alice pub key: $A = \alpha^a \pmod p$

Bob pub key: $B = \alpha^b \pmod p$

Shared key = $K = B^a \pmod p = A^b \pmod p$

5.1 $a = 3$, $b = 5$

$A = 2^3 \pmod{467} = 8$

$B = 2^5 \pmod{467} = 32$

$K = 32^3 \pmod{467} = 78$

$= 8^5 \pmod{467} = 78$

Public Keys = A = 8, B = 32

Common Key = K = 78

5.2 a = 400, b = 134

$A = 2^{400} \pmod{467} = 137$

$B = 2^{134} \pmod{467} = 84$

$K = 84^{400} \pmod{467} = 90$

$= 137^{134} \pmod{467} = 78$

Public Keys = A = 137, B = 84

Common Key = K = 90

5.3 a = 228, b = 57

$A = 2^{288} \pmod{467} = 394$

$B = 2^{57} \pmod{467} = 313$

$K = 313^{228} \pmod{467} = 206$

$= 394^{57} \pmod{467} = 206$

Public Keys = A = 394, B = 313

Common Key = K = 206

6. p = 467, $\alpha = 4$. Order of 4 is 233

So $4^{233} \equiv 1 \pmod{467}$

DH – Shared key

$K_{AB} = a^{ab} \pmod{p}$

α has order 233, we can further reduce the exponent by mod 233

to get $4^{ab} \pmod{467} = 4^{ab \pmod{233}} \pmod{467}$

6.1 $a = 400$, $b = 134$

$ab = 400 * 134 = 53600$, $53600 \pmod{233} = 10$

so, $K_{AB} = 4^{53600} \pmod{467} = 4^{10} \pmod{467}$

$4^1 = 4$

$4^2 = 16$

$4^3 = 64$

$4^4 = 256$

$4^5 = 4 * 4^4 = 4 * 256 = 1024 \equiv 90$

$4^6 = 4 * 90 = 360$

$4^7 = 4 * 4^6 = 4 * 360 = 1440 \equiv 39$

$4^8 = 4 * 39 = 156$

$4^9 = 4 * 4^8 = 4 * 156 = 624 \equiv 157$

$4^{10} = 4 * 4^9 = 4 * 157 = 628 \equiv 161$

So $K_{AB} = 161$

6.2 $a = 167$, $b = 134$

$ab = 167 * 134 = 22378$, $22378 \pmod{233} = 10$

So, $K_{AB} = 4^{22378} \pmod{467} = 4^{10} \pmod{467} = 161$

So, $K_{AB} = 161$

6.3

$400 \equiv 167 \pmod{233}$, since $400 = 167 + 233$.

So, $4^{100} \equiv 4^{167} \pmod{467}$. Public keys $a = 400$ and $a = 167$ are actually the same element in the subgroup. Since $\alpha = 4$ and has order 233, the exponents that differ by a multiple of 233 will produce the same public value and same shared key.

7.

7.1

$p - 1 \equiv -1 \pmod{p}$

$a^1 = p - 1 \equiv -1 != 1 \pmod{p}$

$a^2 = (p - 1)^2 \equiv (-1)^2 = 1 (\mathrm{mod}\ p)$

So, the smallest positive integer k such that $a^k \equiv 1 \pmod p$ is $k = 2$.

7.2

$A = \{a^k \pmod p \mid k \geq 0\}$

Since the order of α is 2, the subgroup will have 2 elements:

$a^0 \equiv 1 \pmod p$

$a^1 \equiv p - 1 \pmod p$

So, $(p - 1) = \{1, p - 1\}$

7.3 If it doesn't use a primitive element but instead an element of a small order then all the DH keys would be in a tiny subgroup. And with the case above the shared key can only be 1 or p – 1, the attacker would simply just have to try both possibilities to break confidentiality. And since the attack can alter values on the channel they could replace one of the public DH messages with a small order element like p – 1. That would force both parties to compute their key inside this two-element subgroup. The attacker can then brute force all possible shared key.

8 Group Z^*_{23}

8.1 $\alpha = 5$, needs to be order 22

$5 \equiv 1 \pmod{23}$

Check

$5^2 = 25 \equiv 2 \pmod{23}$

$5^4 = (5^2)^2 \equiv 2^2 = 4$

$5^8 = (5^4)^2 \equiv 4^2 = 16$

$5^9 = 5 * 5^8 = 5 * 16 = 80 \equiv 11 \pmod{23}$

$5^{10} = 5 * 5^9 = 5 * 11 = 55 \equiv 9 \pmod{23}$

$5^{11} = 5 * 5^{10} = 5 * 9 = 45 \equiv 45 - 46 = -1 \equiv 22 \pmod{23}$

So, $5^{11} \equiv 22 \mathrel{!=} 1 \pmod{23}$

The order of 5 must divide 22 and is not 1, 2 or 11 so it must be

ord(5) = 22, and a generator of Z^*_{23}

8.2 check 2,3,4 if order 22

$2^2 = 4 \mathrel{!=} 1$

$2^4 = 4^2 = 16$

$2^8 = 16^2 = 256 \equiv 3 \pmod{23}$

$2^{11} = 2^8 * 2^2 * 2^1 \equiv 3 * 4 * 2 = 24 \equiv 1 \pmod{23}$

So, ord(2) = 11, not 22

$3^2 = 9$!= 1

$3^4 = 9^2 = 81 \equiv 12 \pmod{23}$

$3^8 = 12^2 = 144 \equiv 6 \pmod{23}$

$3^{11} = 3^8 * 3^2 * 3^1 \equiv 6 * 9 * 3 = 162 \equiv 1 \pmod{23}$

So, ord(3) = 11, not 22

$4^2 = 16$!= 1

$4^4 = 16^2 = 256 \equiv 3 \pmod{23}$

$4^8 = 3^2 = 9$

$4^{11} = 4^8 * 4^2 * 4^1 \equiv 9 * 16 * 4 = 576 \equiv 1 \pmod{23}$

So, ord(4) = 11, not 22

2,3 and 4 all have order 11 not 22, so 5 is the smallest generator of Z^*_{23}

8.3 $\log_5{}^8$ in Z^*_{23} , elements = 22

Find x for $5^x \equiv 8 \pmod{23}$

x = 1

$5^1 = 5 \equiv 5 \pmod{23}$

x = 2

$5^2 = 25 \equiv 2 \pmod{23}$

x = 3

$5^3 = 5 * 5^2 \equiv 5 * 2 = 10 \pmod{23}$

x = 4

$5^4 = 5 * 5^3 \equiv 5 * 10 = 50 = 4 \pmod{23}$

x = 5

$5^5 = 5 * 5^4 \equiv 5 * 4 = 20 \pmod{23}$

x = 6

$5^6 = 5 * 5^5 \equiv 5 * 20 = 100 \equiv 8 \pmod{23}$

$5^6 \equiv 8 \pmod{23}$

$x = 6$

so $\log_5 8 = 6$ in Z^*_{23}

9. is_generator.py

10. computelog.py